



FFI Norwegian Defence
Research Establishment



World-Model-Grounded LLM Planning for AUV and ASV Navigation Near Offshore Wind Farms

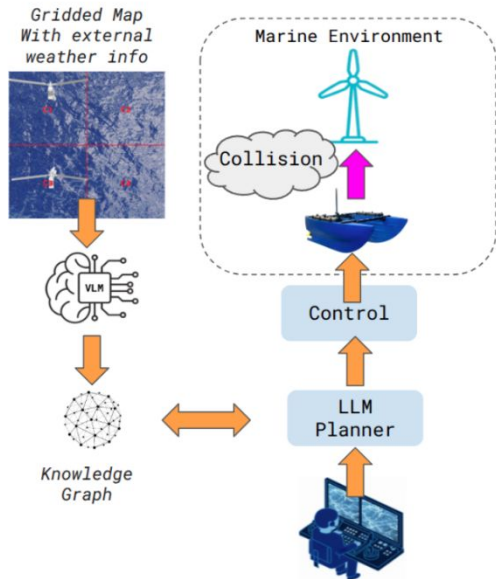
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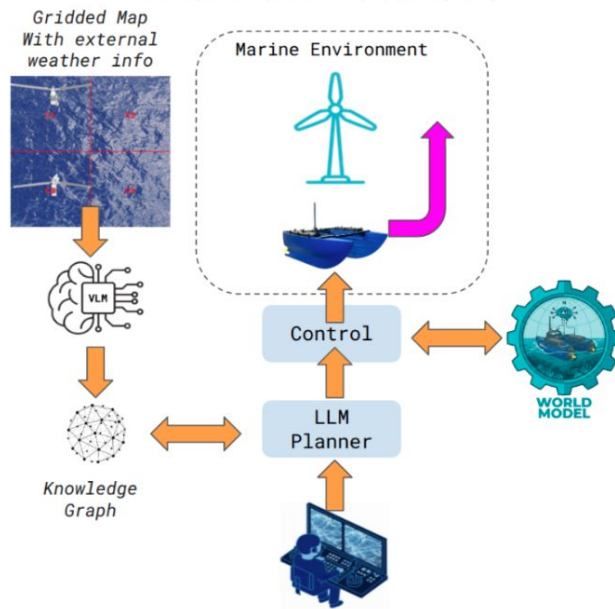
Motivation

NO WORLD MODEL -> Collisions



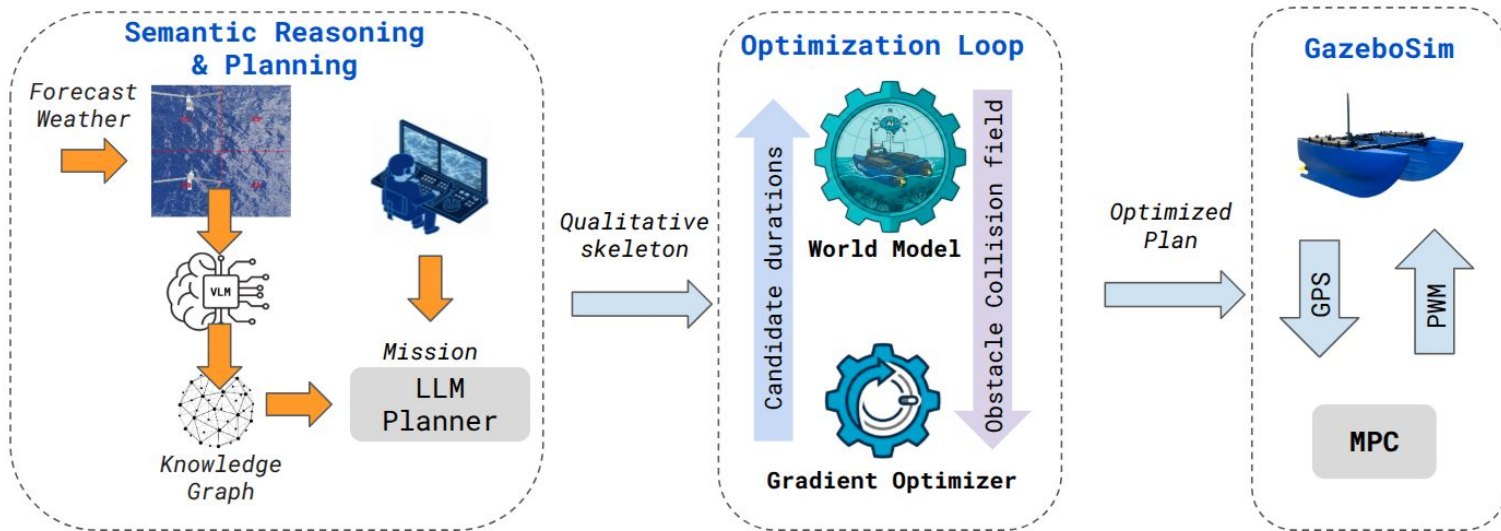
LLMs map missions to actions—**without** understanding physics.

WITH WORLD MODEL -> NO Collisions



Our framework combines an LLM (**what**) and a physics world model (**when**).

The Architecture & Method



1. Physics-Grounded World Model: Combines marine dynamics with learned residuals for 6-DOF AUVs and 3-DOF ASVs.

2. Gradient Trajectory Optimizer: Tunes step durations via differentiable collision fields.

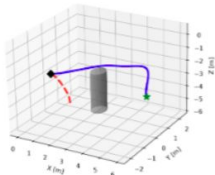
3. Trust-Region MPC: Closed-loop replanning with drift-safe state feedback.

4. VLM Semantic Mapping: Builds obstacle maps from satellite imagery, nautical charts, and weather data.

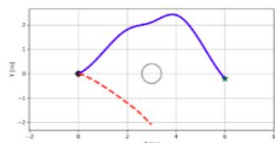
Key Results



GazeboSim (AUV)



AUV 3-D



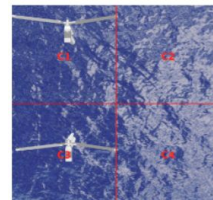
AUV top-down

100% Collision-Free

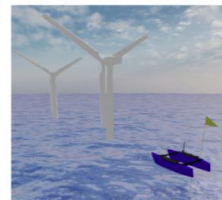
Navigation: Reached all goals in GazeboSim under waves and currents.

Up to 93% Less Error:

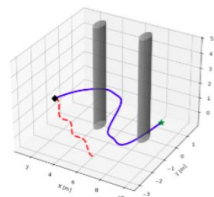
Reduced goal-distance error by 70%–82% (ASV) and ~93% (AUV) versus baseline LLMs.



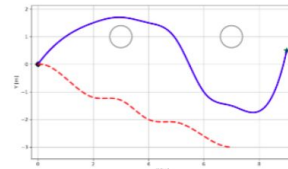
VLM grid cells



GazeboSim (ASV)



ASV 3-D



ASV top-down

96% Map Accuracy: Matched hand-drawn obstacle maps using only open satellite and marine data.